Blood Bank Management System

Software Test Plan

Part B Software Test Plan

Course UE24CS341A Software Engineering
Document identifier BBMS STP 1.0
Version 1.0 29 September 2026

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Document Control

Item	Value
Document title	Software Test Plan
Document identifier	BBMS STP 1.0
Version	1.0
Status	Submission copy
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Course	UE24CS341A Software Engineering
System scope	Verification of donor, donation, screening, inventory and hospital request management

Revision History

Version	Date	Description	Authors
1.0	29 September 2026	Complete part b software test plan for Part 1 submission	Project team

Approval Record

Role	Name	Signature	Date
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Team representative			

Table of Contents

1 Introduction

1.1 Purpose

This Test Plan defines the test scope, approach, environment, responsibilities, acceptance criteria and traceability for the Blood Bank Management System. It covers functional, integration, performance, compatibility, recovery and security validation.

1.2 Test Objectives

- Verify that every high-priority workflow satisfies its linked SRS requirements.
- Verify that invalid input, authorization failures and concurrent operations do not corrupt data.
- Measure the specified performance, compatibility and recovery requirements.
- Provide repeatable evidence for acceptance of the academic prototype.

1.3 References

This plan uses the requirement baseline defined in the Software Requirements Specification, document identifier BBMS SRS 1.0, and the standards listed in the References section of this document.

1.4 Test Items

- Django web application and REST-style endpoints.
- PostgreSQL schema, constraints and transactional behaviour.
- Role-specific browser interfaces.
- Scheduled alert, expiry and backup tasks.
- Email-provider integration through a controlled test service.

2 Test Scope

2.1 Included

Testing includes all functional requirements, security requirements and measurable non-functional requirements that can be evaluated in the academic environment. Unit, integration, system and acceptance-level checks are included.

2.2 Excluded

The plan excludes clinical certification, laboratory-analyser integration, biometric identity verification, national-scale load testing and assessment of physical blood storage conditions. These exclusions do not remove the need for controlled validation before real deployment.

3 Features to Be Tested

Feature	Requirement groups	Primary test level
Donor and appointment management	FR-01 to FR-08	System and acceptance
Donation, screening and unit lifecycle	FR-09 to FR-16	Unit, integration and system

Feature	Requirement groups	Primary test level
Inventory, requests and issue	FR-17 to FR-24	Integration, system and concurrency
Authentication, administration, reports and audit	FR-25 to FR-30	Security and system
Quality requirements	NFR-01 to NFR-12	Performance, recovery, compatibility and review
Security controls	SEC-01 to SEC-10	Security test and code review

3.1 Features Not to Be Tested

Third-party email infrastructure availability and operating-system security are outside direct product testing. Integration tests use a controlled email service. Infrastructure configuration is inspected separately.

4 Test Approach

4.1 Test Levels

Level	Focus	Examples
Unit	Business-rule and state-transition functions	Eligibility, expiry and request transitions
Integration	Database, email and scheduled-task boundaries	Unit release, alerts and transaction rollback
System	End-to-end role workflows	Registration, screening and issue
Acceptance	SRS pass criteria in a representative environment	Performance, recovery and supported browsers

4.2 Test Techniques

- Equivalence partitioning for valid and invalid form input.
- Boundary-value analysis for age, weight, interval, capacity, quantity and time limits.
- State-transition testing for units and blood requests.
- Concurrency testing for appointment capacity and unit reservation.
- Negative authorization tests using direct URLs and API requests.
- Performance tests with measured response percentiles.
- Backup restoration and browser compatibility checks.

4.3 Test Environment

Area	Planned environment
Application	Django application configured with production-like settings except controlled test credentials
Database	PostgreSQL with representative test data and isolated test schema
Browsers	Latest two major versions of Chrome, Edge and Firefox available during acceptance

Area	Planned environment
Email	Local or sandbox SMTP service that records sent messages
Load generation	Scripted HTTP clients producing 100 authenticated concurrent users
Security	Automated dependency and static checks plus manual authorization tests

5 Test Management

5.1 Security Validation

Security validation covers authentication failure handling, account lockout, session expiry, role and branch authorization, CSRF protection, input handling, medical-data exposure, audit completeness and backup access. Tests use synthetic records only. No real donor or patient data is used.

Security area	Validation method	Evidence
Authentication	Failed-login, lockout, logout and session-expiry tests	Request log and audit event
Authorization	Role matrix, direct URL and cross-branch access tests	HTTP result and audit event
Input protection	CSRF and stored-script payload tests	Rejected request or safely encoded display
Audit	Required-field and edit-prevention tests	Exported audit records
Privacy	Hospital and report field inspection	Screenshots and response-body comparison
Backup	Encryption, access restriction and restoration test	Backup log and restored record comparison

5.2 Entry Criteria

- The requirement baseline and database schema are versioned.
- The selected build installs successfully in the test environment.
- Required test accounts and synthetic data are available.
- Critical unit tests pass before system testing begins.

5.3 Exit Criteria

- All listed test cases have an execution result and evidence.
- All high-priority functional and security tests pass.
- No open defect causes data corruption, unauthorized access or incorrect issue of a unit.
- At least 90 percent of all planned test cases pass, with remaining defects documented and accepted by the evaluator.
- Performance and backup-restoration acceptance criteria pass in the defined environment.

5.4 Suspension and Resumption Criteria

Testing is suspended when the build cannot start, the test database is unavailable, a defect corrupts shared test data or more than 20 percent of attempted cases are blocked by one defect. Testing resumes after the cause is fixed, the environment is restored and a smoke test passes.

5.5 Pass and Fail Rules

A test passes only when the observed result matches every stated expected result. A difference in stored data, authorization decision, state transition, response-time limit or displayed result causes failure. Blocked tests are not counted as passed.

5.6 Deliverables

- Approved Test Plan
- Test data set and scripts
- Execution log and evidence
- Defect log
- Test summary report

5.7 Responsibilities

Role	Responsibility
Test coordinator	Maintain plan, environment, execution order and summary
Tester	Prepare data, execute cases, save evidence and report defects
Developer	Support diagnosis, correct defects and provide a new build
Team reviewer	Check traceability and confirm exit criteria

5.8 Risks and Mitigation

Risk	Effect	Mitigation
Limited project time	Reduced regression coverage	Automate unit tests and run high-priority cases first
Unstable external email service	Blocked notification tests	Use a local or sandbox SMTP service
Insufficient representative data	Unrealistic performance result	Generate synthetic data at the stated scale
Shared test records	Tests influence one another	Reset an isolated database before each suite

6 Test Cases

The following nineteen cases exceed the required minimum of seven to ten and cover functional, non-functional and security requirements.

6.1 TC-F01 Donor registration and duplicate prevention

Field	Specification
Requirement links	FR-01, FR-02
Test type	Functional
Preconditions	No active donor exists for donor1@example.test.
Test data	Valid mandatory fields, consent accepted, email donor1@example.test.
Procedure	1. Submit a complete donor registration. 2. Submit the same email address again.
Expected result and pass criterion	The first request creates one donor profile. The second request is rejected with a duplicate-account message and no second profile is stored.

6.2 TC-F02 Eligibility rules and failed-reason display

Field	Specification
Requirement links	FR-03, FR-04
Test type	Functional and boundary
Preconditions	Eligibility rules are configured according to the approved policy.
Test data	Eligible donor, under-age donor, active-deferral donor, and donor inside the configured donation interval.
Procedure	Evaluate eligibility for each test profile and attempt appointment confirmation.
Expected result and pass criterion	Only the eligible profile can confirm an appointment. Each blocked profile receives the applicable reason without disclosure of unrelated medical data.

6.3 TC-F03 Appointment capacity and cancellation

Field	Specification
Requirement links	FR-05, FR-06
Test type	Functional and concurrency
Preconditions	A slot with capacity one exists and two eligible donors are active.
Test data	Two concurrent appointment requests for the same slot.
Procedure	Submit both requests concurrently. Cancel the successful booking and retry the rejected request.
Expected result and pass criterion	Only one initial booking succeeds. After cancellation, the second donor can book the released capacity.

6.4 TC-F04 Donor history access

Field	Specification
Requirement links	FR-07, SEC-01
Test type	Functional and security
Preconditions	Two donors with different records exist.
Test data	Authenticated donor A and record identifiers for donors A and B.
Procedure	Open donor A history. Attempt to request donor B history.
Expected result and pass criterion	Donor A can view only donor A records. Access to donor B records is denied and logged.

6.5 TC-F05 Donor deferral

Field	Specification
Requirement links	FR-08
Test type	Functional
Preconditions	Authorized staff and active donor exist.
Test data	Temporary deferral with valid reason and end date, followed by a permanent deferral without an end date.
Procedure	Record each valid deferral and attempt an invalid temporary deferral without an end date.
Expected result and pass criterion	Valid deferrals are stored with staff identity. The incomplete temporary deferral is rejected. Eligibility reflects the active deferral.

6.6 TC-F06 Donation and unique unit creation

Field	Specification
Requirement links	FR-09, FR-10
Test type	Functional and data integrity
Preconditions	Eligible checked-in donor and authorized staff exist.
Test data	One whole-blood donation with component preparation instructions.
Procedure	Record the collection and generate component units. Attempt to reuse a generated unit ID.
Expected result and pass criterion	The donation and units are stored with unique IDs and complete collection metadata. Duplicate unit ID storage fails.

6.7 TC-F07 Laboratory screening and quarantine

Field	Specification
Requirement links	FR-11, FR-12, FR-13
Test type	Functional and negative
Preconditions	Collected unit exists in Testing state.
Test data	Complete acceptable panel, incomplete panel, and panel containing one reactive result.
Procedure	Submit each result set for separate units and request release to inventory.
Expected result and pass criterion	The acceptable unit becomes Available. Incomplete and reactive units remain unavailable and enter Quarantined state as configured.

6.8 TC-F08 Unit state transitions and expiry

Field	Specification
Requirement links	FR-14, FR-15, FR-16
Test type	Functional and scheduled job
Preconditions	Shelf-life rules exist for the tested component.
Test data	Available unit just before expiry, after expiry, and an invalid direct Collected-to-Issued transition.
Procedure	Run the expiry task and attempt the invalid state transition.
Expected result and pass criterion	The expired unit becomes Expired and disappears from available stock. The invalid transition is rejected and logged.

6.9 TC-F09 Authorized inventory search

Field	Specification
Requirement links	FR-17, NFR-11
Test type	Functional and privacy
Preconditions	Available and unavailable units exist across branches.
Test data	Blood group, component, branch and expiry filters.
Procedure	Search as staff and hospital staff with the same filters.
Expected result and pass criterion	Both roles see permitted availability data. Medical history and screening details are absent. Unavailable units are excluded.

6.10 TC-F10 Low-stock and near-expiry alerts

Field	Specification
Requirement links	FR-18

Field	Specification
Test type	Functional and integration
Preconditions	Thresholds, recipients and near-expiry period are configured.
Test data	Stock count crossing below threshold and unit entering near-expiry period.
Procedure	Run inventory update and scheduled alert task twice.
Expected result and pass criterion	One alert per new condition is sent and recorded. The repeated run does not create a duplicate alert for the unchanged condition.

6.11 TC-F11 Blood request workflow

Field	Specification
Requirement links	FR-19, FR-20
Test type	Functional and workflow
Preconditions	Hospital and staff accounts are active.
Test data	Valid request and invalid request without required-by time.
Procedure	Submit both requests. Approve the valid request and reject another valid request.
Expected result and pass criterion	The incomplete request is rejected. Valid state changes store actor and UTC time and follow the permitted transition rules.

6.12 TC-F12 Request cancellation permissions

Field	Specification
Requirement links	FR-21, SEC-01
Test type	Functional and security
Preconditions	One Pending and one Approved request exist.
Test data	Authenticated hospital staff and blood bank staff accounts.
Procedure	Hospital staff cancel both requests. Blood bank staff cancel the approved request.
Expected result and pass criterion	Hospital staff can cancel only the Pending request. Staff can cancel the Approved request. All attempts are logged.

6.13 TC-F13 Reservation, cross-match and atomic issue

Field	Specification
Requirement links	FR-22, FR-23, FR-24
Test type	Functional, concurrency and integrity
Preconditions	Approved requests and compatible Available units exist.

Field	Specification
Test data	Two concurrent reservations for the same unit, failed cross-match, and successful issue.
Procedure	Run concurrent reservations. Record failed cross-match. Record successful cross-match and issue. Simulate failure during issue commit.
Expected result and pass criterion	One reservation succeeds. Failed cross-match prevents issue. Successful issue updates unit and request atomically. Simulated failure rolls back both updates.

6.14 TC-S01 Login failure, lockout and generic message

Field	Specification
Requirement links	FR-25, SEC-03, SEC-10
Test type	Security
Preconditions	One active account exists.
Test data	Incorrect passwords for registered and unregistered emails.
Procedure	Submit five failed attempts within 10 minutes and then submit correct credentials during the lockout period.
Expected result and pass criterion	Both email cases show the same generic failure message. The account locks for 15 minutes, correct credentials are rejected during lockout, and the event is audited.

6.15 TC-S02 Role-based authorization

Field	Specification
Requirement links	FR-26, SEC-01, NFR-09
Test type	Security
Preconditions	Accounts for all four roles exist.
Test data	Representative URLs and actions for donor, staff, hospital staff and administrator.
Procedure	Attempt each protected action using every role, including a request for another branch outside the assigned scope.
Expected result and pass criterion	Only permissions in the approved role matrix succeed. Direct URL access and out-of-scope branch access are denied and logged.

6.16 TC-S03 Audit completeness and immutability

Field	Specification
Requirement links	FR-30, SEC-07
Test type	Security and audit
Preconditions	Authorized operational and administrator accounts exist.

Field	Specification
Test data	Screening edit, issue, role change and rejected authorization attempt.
Procedure	Perform each event, inspect the audit record, and attempt to edit an audit event through the application.
Expected result and pass criterion	Every event records actor, type, UTC time, record ID and outcome. No normal role can edit or delete an audit event.

6.17 TC-N01 Search and dashboard response time

Field	Specification
Requirement links	NFR-01
Test type	Performance
Preconditions	Representative database contains at least 10,000 units and 5,000 donors.
Test data	100 concurrent authenticated users for 15 minutes.
Procedure	Run representative search and dashboard requests and record server response times.
Expected result and pass criterion	At least 95 percent of measured responses complete within 3 seconds and no request returns an integrity error.

6.18 TC-N03 Encrypted backup and restoration

Field	Specification
Requirement links	NFR-04, SEC-08
Test type	Recovery and security
Preconditions	Backup schedule and restricted storage are configured.
Test data	Known donor, unit, request and audit records.
Procedure	Run backup, verify encryption and access restriction, restore into an empty acceptance environment, and compare known records.
Expected result and pass criterion	The authorized service creates and reads the encrypted backup. Unauthorized access fails. Restoration completes within 4 hours with all known records intact.

6.19 TC-N05 Responsive browser layout

Field	Specification
Requirement links	NFR-06
Test type	Compatibility and usability
Preconditions	Supported browser versions are available.
Test data	Chrome, Edge and Firefox at 360, 768 and 1440 pixel viewport widths.

Field	Specification
Procedure	Open registration, appointment, inventory and request forms at each size and complete one valid submission.
Expected result and pass criterion	No standard form requires horizontal scrolling, labels remain associated with controls, and each submission completes successfully.

7 Test Traceability

Table 1 Test-to-requirement traceability

Test case	Requirement links	Coverage	Initial status
TC-F01	FR-01, FR-02	Functional	Not executed
TC-F02	FR-03, FR-04	Functional and boundary	Not executed
TC-F03	FR-05, FR-06	Functional and concurrency	Not executed
TC-F04	FR-07, SEC-01	Functional and security	Not executed
TC-F05	FR-08	Functional	Not executed
TC-F06	FR-09, FR-10	Functional and data integrity	Not executed
TC-F07	FR-11, FR-12, FR-13	Functional and negative	Not executed
TC-F08	FR-14, FR-15, FR-16	Functional and scheduled job	Not executed
TC-F09	FR-17, NFR-11	Functional and privacy	Not executed
TC-F10	FR-18	Functional and integration	Not executed
TC-F11	FR-19, FR-20	Functional and workflow	Not executed
TC-F12	FR-21, SEC-01	Functional and security	Not executed
TC-F13	FR-22, FR-23, FR-24	Functional, concurrency and integrity	Not executed
TC-S01	FR-25, SEC-03, SEC-10	Security	Not executed
TC-S02	FR-26, SEC-01, NFR-09	Security	Not executed
TC-S03	FR-30, SEC-07	Security and audit	Not executed
TC-N01	NFR-01	Performance	Not executed
TC-N03	NFR-04, SEC-08	Recovery and security	Not executed
TC-N05	NFR-06	Compatibility and usability	Not executed

8 Test Schedule

Activity	Planned sequence	Output
Unit and integration preparation	1	Automated checks and test data
Functional system testing	2	Functional execution results

Activity	Planned sequence	Output
Security and recovery testing	3	Security and restoration evidence
Performance and compatibility testing	4	Measurements and browser evidence
Regression and summary	5	Final test summary and defect status

References

[1] ISO IEC IEEE 29148 2018, Systems and software engineering, Life cycle processes, Requirements engineering.

[2] IEEE 829 2008, Standard for Software and System Test Documentation, used as the course-aligned document structure. The standard has been superseded by the ISO IEC IEEE 29119 series.

[3] IEEE 1016 2009, Standard for Information Technology, Systems Design, Software Design Descriptions.

[4] Ministry of Health and Family Welfare, Government of India, National Standards for Blood Centres, Annexure 4 Criteria for Selection of Blood Donor.

[5] Central Drugs Standard Control Organization, Drugs and Cosmetics Act 1940 and Rules 1945, Schedule F provisions for blood banks.

[6] eRaktKosh, Ministry of Health and Family Welfare, overview of centralized donor, testing, inventory and notification functions.

[7] Django Software Foundation, Django documentation for authentication, security and database transactions.

Appendix A Compliance Checklist

Deliverable requirement	Location in this document	Status
Test Plan in IEEE-aligned format	Sections 1 to 8	Complete
Test Plan Sections 3, 4 and 5	Sections 3 to 5	Complete
Security validation in Section 5.1	Section 5.1	Complete
Traceability with the SRS	Section 7	Complete
At least seven to ten functional and non-functional test cases	Section 6 contains nineteen cases	Complete